

## YANJI, China

WMO: 542920

Lat: 42.8737N

Lon: 129.5045E

Elev: 258

StdP: 98.26

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |       |      |       | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|-------|------|-------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |       | 1%   |       |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB  | WS   | MCDB  | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)   | (l)  | (m)   | (n)                   | (o)  | (p)   |
| 1             | -21.8      | -19.8 | -29.0                         | 0.3 | -18.2 | -27.3 | 0.3 | -17.0 | 12.0                  | -13.1 | 11.0 | -12.8 | 1.8                   | 270  | 0.509 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 8.5                    | 31.4            | 21.8 | 29.7 | 21.0 | 28.1 | 20.4 | 24.2                | 28.9 | 23.1 | 26.9 | 22.2 | 25.8 | 2.9                  | 290  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 22.8                            | 18.0 | 26.9 | 21.9 | 17.1 | 25.7 | 21.0 | 16.2 | 24.5 | 74.8          | 29.0 | 70.4 | 26.8 | 66.8 | 25.8 | 27.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 10.5              | 9.1  | 7.8 | -23.5                      | 34.2 | 2.5                | 1.3 | -25.2                                              | 35.2 | -26.7      | 36.0 | -28.1      | 36.7 | -29.9      | 37.7 |    |    |
|                   |      |     | -23.9                      | 25.4 | 2.4                | 1.1 | -25.6                                              | 26.2 | -26.9      | 26.9 | -28.3      | 27.5 | -30.0      | 28.4 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 6.3    | -12.2 | -8.1  | -0.9  | 7.6   | 14.0  | 18.8  | 22.1  | 21.9  | 15.9  | 7.9   | -2.3  | -10.2 |
|                                                                                  | DBStd       | 12.73  | 3.53  | 4.42  | 4.75  | 4.31  | 3.85  | 3.48  | 3.07  | 2.67  | 3.23  | 4.05  | 5.18  | 3.99  |
|                                                                                  | HDD10.0     | 2725   | 689   | 508   | 337   | 97    | 9     | 0     | 0     | 2     | 90    | 368   | 626   |       |
|                                                                                  | HDD18.3     | 4702   | 948   | 741   | 595   | 322   | 142   | 36    | 5     | 84    | 324   | 618   | 884   |       |
|                                                                                  | CDD10.0     | 1373   | 0     | 0     | 1     | 26    | 133   | 263   | 377   | 369   | 179   | 24    | 0     | 0     |
|                                                                                  | CDD18.3     | 308    | 0     | 0     | 0     | 1     | 8     | 49    | 123   | 114   | 12    | 0     | 0     | 0     |
|                                                                                  | CDH23.3     | 2296   | 0     | 0     | 0     | 22    | 171   | 437   | 878   | 699   | 88    | 2     | 0     | 0     |
|                                                                                  | CDH26.7     | 671    | 0     | 0     | 0     | 4     | 48    | 136   | 294   | 181   | 9     | 0     | 0     | 0     |
| Wind                                                                             | WSAvg       | 2.6    | 3.0   | 3.0   | 3.2   | 3.2   | 2.9   | 2.5   | 2.2   | 2.0   | 1.8   | 2.3   | 2.5   | 2.8   |
| Precipitation                                                                    | PrecAvg     | 515    | 4     | 6     | 10    | 28    | 55    | 86    | 112   | 125   | 56    | 26    | 15    | 6     |
|                                                                                  | PrecMax     | 736    | 23    | 28    | 42    | 76    | 129   | 338   | 214   | 289   | 204   | 112   | 79    | 51    |
|                                                                                  | PrecMin     | 309    | 0     | 0     | 0     | 2     | 7     | 21    | 16    | 13    | 4     | 1     | 1     | 0     |
|                                                                                  | PrecStd     | 107    | 4     | 6     | 8     | 17    | 33    | 48    | 48    | 69    | 44    | 20    | 15    | 7     |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 0.0   | 7.3   | 17.3  | 26.1  | 31.0  | 32.7  | 34.0  | 32.6  | 28.0  | 22.3  | 14.3  | 2.7   |
|                                                                                  |             | MCWB   | -3.3  | 1.3   | 8.1   | 12.8  | 16.3  | 20.1  | 24.0  | 24.0  | 20.1  | 13.8  | 7.4   | -0.6  |
|                                                                                  | 2%          | DB     | -2.7  | 4.3   | 12.6  | 22.7  | 27.8  | 30.1  | 31.6  | 30.5  | 25.7  | 19.7  | 10.9  | 0.3   |
|                                                                                  |             | MCWB   | -5.6  | -0.6  | 4.7   | 10.9  | 15.0  | 19.3  | 23.2  | 22.8  | 18.2  | 12.3  | 5.5   | -2.6  |
|                                                                                  | 5%          | DB     | -4.5  | 1.8   | 9.9   | 19.5  | 24.9  | 28.0  | 29.9  | 28.7  | 24.0  | 17.5  | 7.9   | -1.9  |
|                                                                                  |             | MCWB   | -7.0  | -2.0  | 3.3   | 9.2   | 14.2  | 18.6  | 22.1  | 21.9  | 17.1  | 10.8  | 3.4   | -4.4  |
|                                                                                  | 10%         | DB     | -6.3  | -0.2  | 7.2   | 16.7  | 22.3  | 25.8  | 28.1  | 27.0  | 22.4  | 15.4  | 5.3   | -3.6  |
|                                                                                  |             | MCWB   | -8.4  | -3.2  | 1.7   | 7.8   | 12.9  | 17.9  | 21.7  | 21.2  | 16.2  | 9.8   | 1.8   | -5.9  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | -3.1  | 1.9   | 8.0   | 14.4  | 18.2  | 22.1  | 26.0  | 25.2  | 21.6  | 15.2  | 8.2   | -0.2  |
|                                                                                  |             | MCDB   | -0.4  | 6.1   | 16.3  | 23.4  | 27.3  | 28.7  | 32.2  | 30.9  | 25.9  | 19.8  | 13.1  | 1.7   |
|                                                                                  | 2%          | WB     | -5.4  | 0.0   | 5.4   | 11.5  | 16.2  | 20.9  | 24.7  | 24.1  | 20.0  | 13.6  | 5.9   | -2.4  |
|                                                                                  |             | MCDB   | -3.0  | 3.5   | 11.5  | 20.5  | 24.6  | 27.0  | 29.7  | 28.6  | 23.5  | 17.9  | 9.8   | 0.0   |
|                                                                                  | 5%          | WB     | -7.0  | -1.7  | 3.6   | 10.0  | 14.9  | 19.8  | 23.6  | 23.2  | 18.6  | 12.3  | 4.0   | -4.3  |
|                                                                                  |             | MCDB   | -4.7  | 1.6   | 9.2   | 18.5  | 22.7  | 25.5  | 27.7  | 26.8  | 22.0  | 15.9  | 7.2   | -2.0  |
|                                                                                  | 10%         | WB     | -8.3  | -3.3  | 2.1   | 8.4   | 13.9  | 18.8  | 22.5  | 22.3  | 17.5  | 10.7  | 2.0   | -5.8  |
|                                                                                  |             | MCDB   | -6.3  | -0.3  | 6.7   | 15.4  | 20.8  | 24.0  | 26.2  | 25.5  | 20.8  | 14.2  | 5.1   | -3.8  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 9.9   | 11.3  | 11.3  | 12.9  | 12.0  | 10.0  | 8.5   | 8.7   | 12.0  | 12.6  | 10.0  | 9.3   |
|                                                                                  |             | MCDBR  | 12.4  | 14.0  | 15.7  | 18.7  | 17.6  | 14.7  | 12.2  | 11.1  | 13.6  | 15.6  | 14.2  | 11.4  |
|                                                                                  |             | MCWBR  | 10.0  | 10.0  | 9.1   | 9.0   | 7.0   | 5.4   | 4.5   | 4.4   | 6.4   | 8.4   | 9.1   | 9.1   |
|                                                                                  | 5% WB       | MCDBR  | 12.1  | 13.3  | 14.5  | 17.3  | 14.9  | 12.1  | 10.1  | 8.9   | 10.7  | 13.3  | 13.4  | 11.1  |
|                                                                                  |             | MCWBR  | 9.9   | 9.7   | 8.9   | 8.9   | 6.6   | 5.1   | 4.5   | 4.1   | 5.6   | 7.9   | 9.1   | 9.0   |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.329  | 0.383 | 0.457 | 0.540 | 0.579 | 0.573 | 0.536 | 0.484 | 0.432 | 0.443 | 0.386 | 0.339 |       |
|                                                                                  | taud        | 2.124  | 1.997 | 1.864 | 1.743 | 1.723 | 1.822 | 1.983 | 2.124 | 2.177 | 2.005 | 2.056 | 2.099 |       |
|                                                                                  | Ebn at Noon | 774    | 782   | 767   | 735   | 721   | 727   | 750   | 779   | 791   | 707   | 689   | 713   |       |
|                                                                                  | Edh at Noon | 102    | 138   | 179   | 218   | 228   | 208   | 175   | 147   | 130   | 136   | 109   | 96    |       |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 2.38   | 3.39  | 4.24  | 4.75  | 5.13  | 5.23  | 4.67  | 4.59  | 4.28  | 3.25  | 2.23  | 1.95  |       |
|                                                                                  | RadStd      | 0.16   | 0.13  | 0.28  | 0.36  | 0.49  | 0.63  | 0.56  | 0.39  | 0.37  | 0.24  | 0.21  | 0.13  |       |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (47) | Regional (10 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page